Ahmedabad University

CSE623 Machine Learning

# Weekly Report 3

**Feature-Based Global Wheat Full Semantic Segmentation using Classical Machine Learning and Contour Analysis**

Submitted to faculty: Prof. Mehul Raval

Date of Submission: 07-03-2026

Student Details

| Roll No. | Name of the Student |
|---|---|
| AU2544013 | Hardi Makwana |
| AU2544022 | Lakshita Rathod |
| AU2544028 | Siddhi Hirani |
| AU2544030 | Krisha Doshi |

2026-2027 (Winter Semester)

## Aim:

To perform a review on semantic segmentation techniques used in wheat head detection and to begin basic Exploratory Data Analysis (EDA) on the Global Wheat Dataset.

## Introduction:

Wheat segmentation plays an important role in improving crop yield and helping global food security, as wheat is a staple food for a large portion of the world's population. Correct detection and segmentation of wheat heads are essential for estimating crop productivity and assisting agronomists in field monitoring.

The Global Wheat Head Dataset (GWHD) provides a large collection of high-resolution field images captured across different environments, making it a valuable resource for wheat head detection research.

## Work Completed:

*Reference Papers Background Study*:

- **Active Contour Models (Kass et al., 1988):** We studied the "Snakes" model to get more idea about preserving boundary detection.In this paper a snake is a flexible curve that automatically adjusts itself by changing its shape to find the edges and boundaries of objects and perfectly fit around the object. In our wheat project, we will use this logic to refine the irregular shapes of wheat spikes and properly separate overlapping heads.
- **GWHD Dataset (David et al., 2020):** We studied the Global Wheat Head Detection dataset to understand the real-world difficulties that wheat is hard to detect as it looks different in every country ,color changes as it develops more and the blurriness due to wind.A major problem is the difficulty of overlapping heads which makes it hard to see where one wheat head ends and another begins.

*Literature Review:*

- Studied the importance of wheat head detection for yield estimation and its role in global food security.

- Reviewed several challenges associated with field-based wheat head detection, including wind-induced blur, varying illumination conditions, and dense overlapping of wheat heads.

- Looked at deep learning approaches such as Mask R-CNN, which achieve high detection accuracy but often lack interpretability because of their black-box nature.

- Also looked at classical machine learning approaches that use hand-crafted features and interpretable models, such as Twin Support Vector Machines (TWSVM), for wheat head detection.

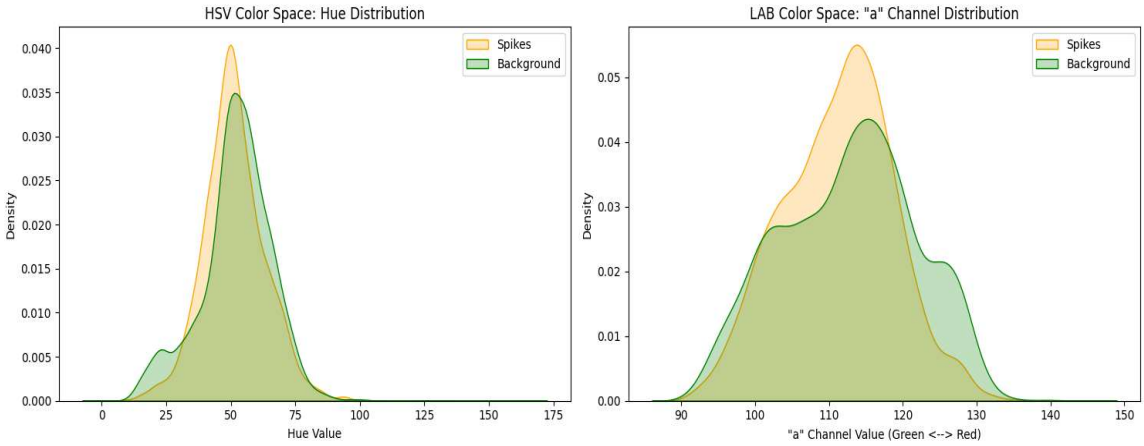
- Studied recent approaches addressing the challenge of manual pixel-level annotation, including the use of synthetic dataset generation techniques such as CycleGAN to create realistic training samples. And also reviewed contour-based and edge-based image processing techniques used for improving the localization of wheat heads and reducing environmental noise in field imagery.

*EDA:*

- We perform the following EDA on the GWHD Dataset:

1. Color Space Separability: We analyze pixel distribution across HSV and LAB color spaces to decide the best channels for isolating wheat spikes from background.

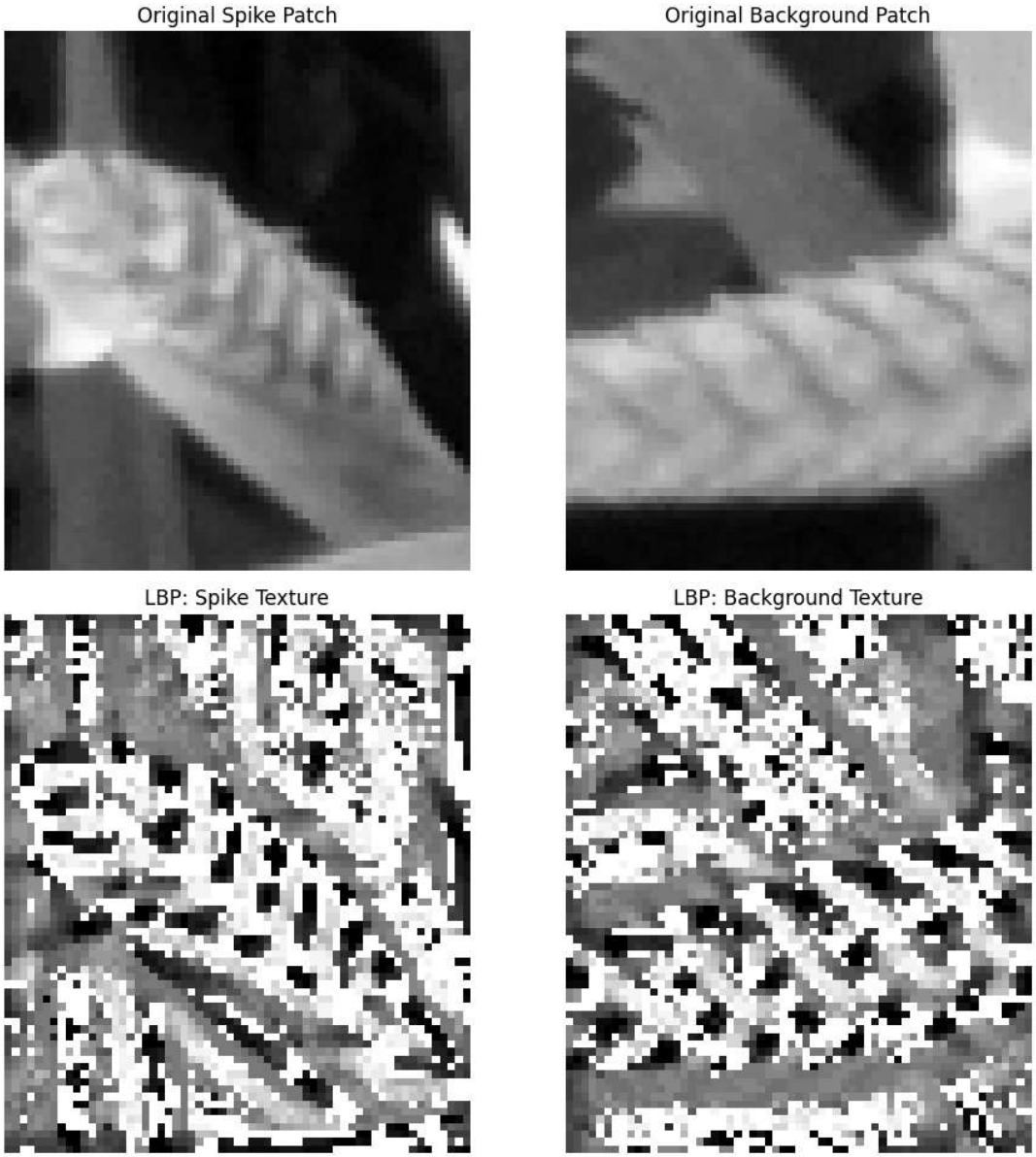
RESULT:




2. Texture Descriptor Analysis: We extract texture features Local Binary Patterns (LBP) and Gray-Level Co-occurrence matrices (GLCM) from different plant organ patches to capture structural differences.

   RESULT:

   A) Local Binary Patterns:




B) Gray-Level Co-occurrence matrices:

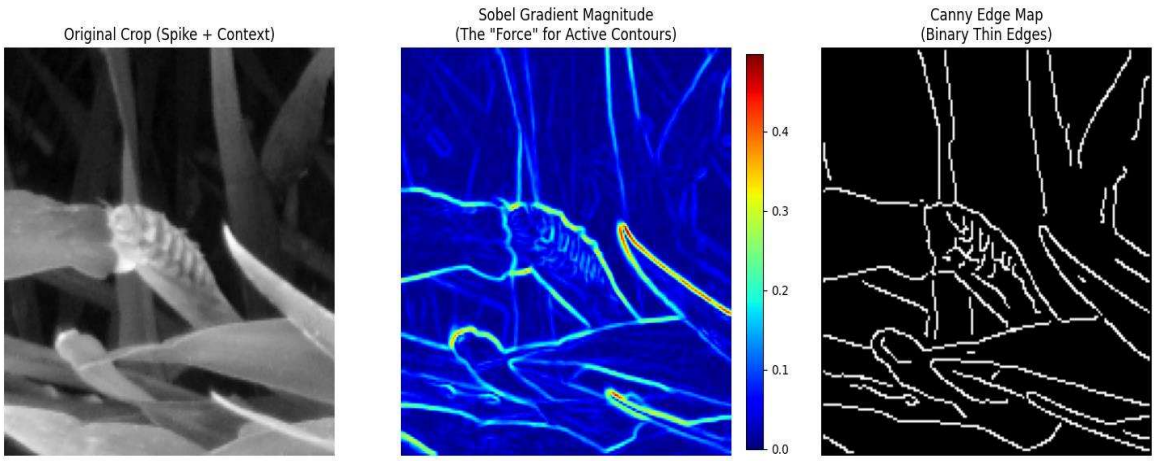
```
Calculating GLCM Texture Signatures...

--- 🌾 SPIKE GLCM Features ---
Contrast:    262.19 (Higher = more rugged/sharp edges)
Energy:      0.0249 (Higher = more uniform/smooth)
Homogeneity: 0.1945 (Higher = fewer texture changes)
Entropy:     46.22 (Higher = more chaotic/complex pattern)

--- 🍃 BACKGROUND GLCM Features ---
Contrast:    163.00 (Higher = more rugged/sharp edges)
Energy:      0.0231 (Higher = more uniform/smooth)
Homogeneity: 0.2041 (Higher = fewer texture changes)
Entropy:     46.12 (Higher = more chaotic/complex pattern)
```

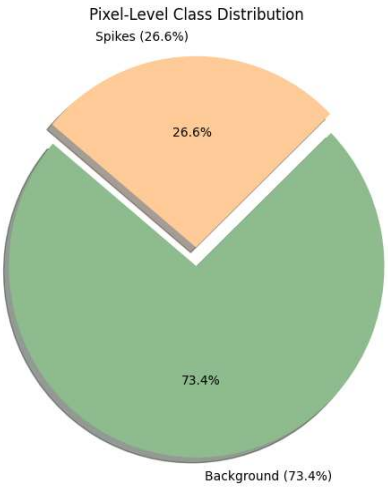
3. Edge Density and Gradient Magnitude: We applied Sobel and Canny edge detectors to evaluate the image energy environment, helps to evaluate the boundaries potential for Active Contour fitting.

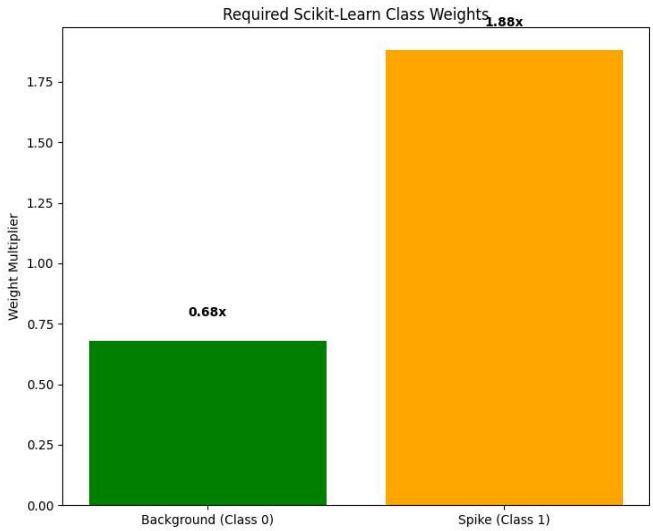
   RESULT:




4. Pixel-Level Class Distribution: Calculating the ratio of "Spike" pixels to "Background" pixels across the dataset to quantify the class imbalance.

   RESULT:




## Next steps and goals:

- Next week we will perform data preprocessing – standardize input images to uniform resolution. They are of same resolution but to ensure all images are of same size we will do that. Then apply gaussian blurring to reduce noise, which will help in edge detection. Instead of using grayscale processing like the traditional method we will preserve 3-channel RGB information that will allow us to leverage colour-based features, which helps to differentiate between green wheat components like spikes and leaves from soil and background.

- And we will implement feature engineering to extract features to use in the classical ML model.

## Conclusion:

This week's activities were dedicated to laying solid groundwork for the wheat head semantic segmentation project by carrying out a series of activities such as literature study and exploratory data analysis. The relevant research articles were studied to get a clear insight into the classical as well as deep learning-based methods for wheat head detection, including the issues faced during the detection of wheat heads from natural scenes such as the presence of overlapping wheat heads, changes in lighting conditions, and the presence of motion blur. Additionally, exploratory data analysis was performed on the Global Wheat Head Dataset to understand the color separability, texture information, edge information, and class information at the pixel level, which are very useful for the development of a feature-based segmentation method using classical machine learning and contour-based methods in the later stages of the project.

## References:


1. David et al. (2020): Global Wheat Head Dataset: A Large and Diverse Dataset of High-Resolution RGB-Images to Develop and Benchmark Wheat Head Detection Methods. https://www.sciencedirect.com/science/article/pii/S2643651524000359?via%3Dihub

2. Kass, M., et al. (1988): Snakes: Active contour models. (Standard for geometric boundary refinement). https://link.springer.com/article/10.1007/bf00133570

3. Wen, Changji, et al. "A generalized model for accurate wheat spike detection and counting in complex scenarios." Scientific reports 14.1 (2024): 24189. https://www.nature.com/articles/s41598-024-75523-w

4. Kalluri, Ramadevi, and Prabha Selvaraj. "An image analysis technique for wheat head count detection using machine learning." Bulletin of Electrical Engineering and Informatics 13.5 (2024): 3515-3525. https://beei2.org/index.php/EEI/article/view/7255

5. David, Etienne, et al. "Global wheat head detection (GWHD) dataset: A large and diverse dataset of high-resolution RGB-labelled images to develop and benchmark wheat head detection methods." Plant Phenomics (2020). https://spj.science.org/doi/full/10.34133/2020/3521852?doi=10.34133/2020/3521852

6. Myers, Jaden, et al. "Modified cyclegan for the synthesization of samples for wheat head segmentation." arXiv preprint arXiv:2402.15135 (2024). https://arxiv.org/abs/2402.15135

7. Sun, Xiaoyong, et al. "Counting wheat heads using a simulation model." Computers and Electronics in Agriculture 228 (2025): 109633. https://www.sciencedirect.com/science/article/pii/S016816992401024X